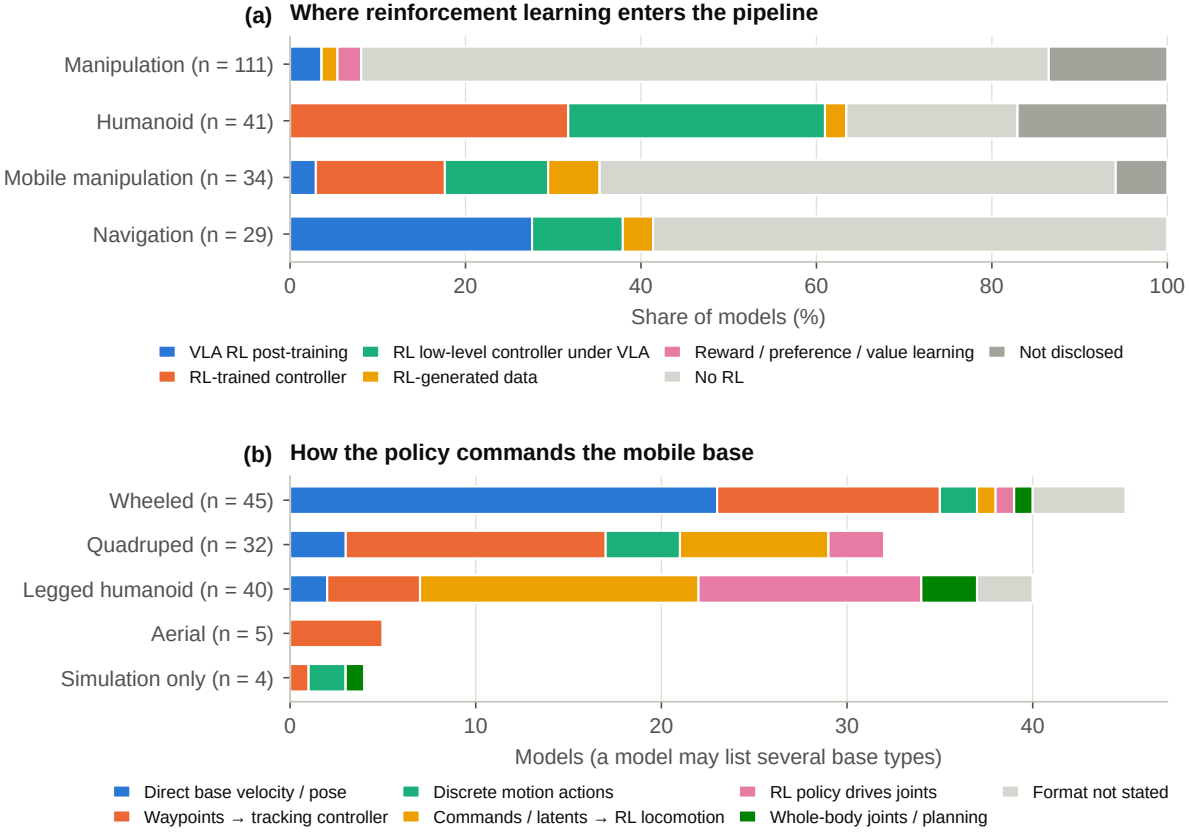
(a) Where reinforcement learning enters the pipeline
Manipulation (n = 111)
Humanoid (n = 41)
Mobile manipulation (n = 34)
Navigation (n = 29)
0
20
40
60
80
100
Share of models (%)
VLA RL post-training
RL low-level controller under VLA
Reward / preference / value learning
Not disclosed
RL-trained controller
RL-generated data
No RL
(b) How the policy commands the mobile base
Wheeled (n = 45)
Quadruped (n = 32)
Legged humanoid (n = 40)
Aerial (n = 5)
Simulation only (n = 4)
0
10
20
30
40
Models (a model may list several base types)
Direct base velocity / pose
Discrete motion actions
RL policy drives joints
Format not stated
Waypoints → tracking controller
Commands / latents → RL locomotion
Whole-body joints / planning